

Thyroid Cheat Sheet

UCSF endocrine modules (Rushakoff, Rao) · ADA 2022–23 era, verify drug details · Pre-op med/insulin dosing tool: [ucsf.logicnets.com](https://www.ucsf.edu/ucsf-logicnets)

RULES

- **Screen: TSH.** + FT4 if suspicious/central · + FT3 if TSH low + FT4 normal · order **free** hormones
- **Ask about biotin** (TSH falsely ↓, FT4 falsely ↑ → fake hyperthyroid). Hold ≥2 d
- **Don't check TFTs in the acutely ill** (sick euthyroid; recovery TSH ↑ is transient)
- **On MMI: fever / sore throat → stop drug, check CBC** (agranulocytosis)

Reading TFTs

TSH	FT4	Means
↑	↓	Primary hypothyroid
↑	nl	Subclinical hypo → repeat in ~2 mo (50% normalize if TSH <8)
↓ / nl	↓	Central hypo (↓FT3 too)
↓	↑	Thyrotoxicosis
↓	nl	Subclinical hyper (check FT3: ↑ = T3 toxicosis)
nl / ↑	↑	TSH-oma vs hormone resistance
nl / sl ↓	nl / ↓ (↓FT3)	Sick euthyroid
↑	↑	LT4 taken right before draw after missed doses

- Steroids, dopamine ↓ TSH · amiodarone/contrast → hypo or hyper · lithium, checkpoint inhibitors → hypo

Hypothyroid

Patient	Levothyroxine start
Young, no cardiac dz, post-thyroidectomy	Full: 1.6 mcg/kg/d
Mild hypo	25–50 mcg/d
Elderly or CAD	12.5 mcg/d , go slow

- **Recheck TSH 6–8 wk** after any change · target normal TSH · don't chase small deviations
- Empty stomach, 1 h before food · **Ca/Fe 4 h apart** · PPI, soy ↓ absorption · missed dose → double next day
- **Subclinical: treat if pregnant**, ↑LDL, or diastolic HTN · else TSH q6–12 mo, treat if >10
- **Pregnant: ↑ dose ~30% right away** (100 → 125)

Hyperthyroid

- **Workup:** TSH, FT4/FT3 → **TRAb/TSI** → **RAIU + scan** if unclear
- **Propranolol 20–40 mg TID** → HR 60–90 for all; taper when euthyroid

Cause	Clues	Treat
Graves	TRAb+, ↑ diffuse uptake, eye signs	MMI (or RAI / surgery)
Toxic MNG / adenoma	Hot nodule(s) on scan	RAI (adenoma preferred) · MMI 10–20 · surgery
Thyroiditis (painful/painless/postpartum)	Uptake ~0 · ↑ESR if painful	β-blocker only, no MMI · NSAIDs or prednisone taper for pain
Exogenous	Uptake ~0, ↓ thyroglobulin	Stop / ↓ LT4

- **MMI 5–20 mg daily** (10 usually enough) · **PTU 50–150 mg TID** only 1st trimester
- Baseline CBC + LFTs · **follow FT4 q3–4 wk, not TSH** (stays suppressed for months) · Rx 1–2 mo at a time
- Stop/↓ MMI if alk phos >3× ULN · contraception (teratogen)
- **Subclinical hyper:** young + asx → observe; risks AF + bone loss in elderly · **pregnant + suspected Graves → endo now**

Nodules

- **TSH first** → low: RAI scan (hot nodule = benign, no FNA) · else **neck US** (no CT, Tg, TPO)
- **FNA:** TR3 ≥2.5 cm · TR4 ≥1.5 cm · TR5 ≥1 cm (UCSF: consider TR4/5 >1 cm)
- **Follow-up US:** TR3 ≥1.5 · TR4 ≥1 · TR5 ≥0.5 cm → TR5 in 6–12 mo, others 12–24 mo; stop after 3–5 yr stable
- **Cystic lymph node, or hard / fast-growing / painful nodule → urgent endo**

Bethesda	Risk	Next
I Nondiagnostic	5–10%	Repeat FNA
II Benign	0–3%	US follow-up
III AUS · IV Follicular	6–40%	Repeat FNA + molecular, or lobectomy
V Suspicious · VI Malignant	45–96%	Surgery